

Benjamin Atawiah
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Department of Mathematics
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RESEARCH INTERESTS

Numerical Analysis and Scientific Computing for Partial Differential Equations

- Discontinuous Galerkin finite element methods (DG-FEM)
- Runge–Kutta time integration and strong stability preserving schemes
- Exponential and implicit–explicit time marching methods
- Stability and error analysis for high-order numerical methods
- Fourier analysis and energy-based stability techniques

EDUCATION

The University of Alabama <i>Ph.D. in Mathematics</i>	Expected Aug. 2026 <i>Advisor: Dr. Zheng Sun</i>
The University of Alabama <i>M.A. in Mathematics</i>	Apr. 2025 <i>Advisor: Dr. Zheng Sun</i>
African Institute for Mathematical Sciences, Ghana <i>M.Sc. in Mathematical Sciences</i>	Jun. 2020 <i>Advisor: Dr. Mohamed Mbehoul</i>
University of Ghana <i>B.A. in Economics and Mathematics</i>	Jun. 2018 <i>Advisor: Mr. Antwi Asare</i>

RESEARCH EXPERIENCE

The University of Alabama	Fall 2024 – Present Graduate Research Assistant (Advisor: Dr. Zheng Sun)
• Investigate stability and accuracy of high-order numerical methods for PDEs. • Perform Fourier analysis for filtered Runge–Kutta discontinuous Galerkin methods with local time marching. • Develop energy-based stability and error estimates for stage-dependent RKDG methods. • Study exponential time differencing methods for dispersive equations with higher-order derivatives. • Explore compact stencil formulations for local discontinuous Galerkin methods to improve computational efficiency.	
African Institute for Mathematical Sciences (AIMS), Ghana	May 2020 – Jun. 2020 Master’s Thesis Research
• Analyzed strange attractors in a climate-related mathematical model. • Applied nonlinear dynamics techniques to study long-term solution stability.	

PUBLICATIONS

1. Atawiah, B., and Sun, Z. *Stability and accuracy effects of filtering on Runge–Kutta discontinuous Galerkin methods: Fourier analysis for the second-order case*. Submitted.
2. Atawiah, B., and Sun, Z. *Energy-based stability and error estimates of stage-dependent Class A Runge–Kutta discontinuous Galerkin methods*. Manuscript in preparation.
3. Atawiah, B., and Sun, Z. *Stability analysis of exponential time differencing with local discontinuous Galerkin methods for dispersive equations with generalized even-order spatial derivatives*. Work in progress.
4. Atawiah, B., and Sun, Z. *Local discontinuous Galerkin methods with compact stencils*. Work in progress.

CONFERENCE PRESENTATIONS

- SIAM Southeast Atlantic Section (SEAS) Conference, Univ. of Tennessee, Knoxville — Oral, Mar. 21–23, 2025.
- Southeast Applied and Computational Mathematics Student Workshop — Oral, Apr. 5–6, 2025.
- Third Joint Alabama–Florida Conference on Differential Equations, UAB — Poster, May 20–22, 2025.
- Winter School: Reduced-Order Modeling for Complex Engineering Problems, IMSI, Univ. of Chicago — Jan. 29–31, 2025.
- Mathematical Modeling of Biological Interfacial Phenomena, IMSI, Univ. of Chicago (Online) — Dec. 9–13, 2024.

TEACHING EXPERIENCE

The University of Alabama

Instructor of Record

Aug. 2022 – Present

- MATH 238 – Applied Differential Equations I (Spring 2024, 2025, Fall 2025, Spring 2026)
- MATH 126 – Calculus II (Fall 2024)
- MATH 227 – Calculus III (Fall 2023)
- MATH 125 – Calculus I (Spring 2023)
- MATH 115 – Algebra and Trigonometry (Fall 2022)

Tutor

Jan. 2022 – Present

- Calculus I–III (Summers 2022–2024)

University of Ghana

Aug. 2018 – Jul. 2019

- Teaching Assistant: MATH 123 – Vectors and Geometry
- Tutor: Calculus I

HONORS AND AWARDS

- Summer Research Fellowship, Dept. of Mathematics, Univ. of Alabama — Summer 2025
- Graduate Student Travel Award, Graduate School, Univ. of Alabama — Mar. 2025
- Travel Awards: SIAM SEAS Conference; SE Applied and Computational Mathematics Student Workshop — 2025
- Graduate Student Travel Funding, JAF DEDS Conference — May 2025
- Edith & Richard Ainsworth Endowed Scholarship, Univ. of Alabama — Summer 2022, 2023
- Full Scholarship, IMSI Winter School on Reduced Order Modeling — Jan. 2025
- AIMS Ghana Fellowship — \$25,000 (Aug. 2019 – Jun. 2020)

COMMUNITY INVOLVEMENT

- Judge, SIMODE (SCUDEM X 2025)
- Judge, SIAM MathWorks Math Modeling (M3) Challenge
- Research Judge, Undergraduate Research and Creative Activity Conference
- Volunteer Grader, MathCounts Competition (2022–2024)
- Event Support, SEC Geometry–Topology Workshop, Univ. of Alabama
- Mathematics Tutor, Korleman Educational Outreach, Ghana

PROGRAMMING SKILLS

Python, MATLAB, Mathematica, Firedrake

PROFESSIONAL MEMBERSHIPS

- American Mathematical Society (AMS), 2022–Present
- Society for Industrial and Applied Mathematics (SIAM), 2022–Present